Predicting Parent-Pulse Multiplicity in Synthetic Gamma-Ray Burst Light Curves with a 1D Convolutional Neural Network

Abstract

Gamma-ray burst (GRB) light curves are highly variable, multi-peaked time series whose substructure — individual "pulses" — is thought to trace discrete episodes of central-engine activity. Counting how many such pulses seeded a given burst is difficult from real data alone: pulses overlap, fade into instrumental noise, and, more fundamentally, no observed burst carries a verifiable ground-truth pulse count, since that quantity is a construct of a generative model rather than a directly measurable one. We address this by training a one-dimensional convolutional neural network (CNN) to predict the number of parent pulses in light curves synthesized from the Stern & Svensson (1996) stochastic pulse-avalanche model, using parameters calibrated to the Swift/BAT instrument by Bazzanini et al. (2024). Across two dataset sizes (3,000 and 10,000 light curves) and three label-distribution regimes — the naturally imbalanced calibrated distribution, a deliberately shifted distribution, and a distribution made uniform by construction — we compare two loss functions and three class-imbalance mitigation strategies. We find that naive training collapses onto whichever pulse count is most common in the training data regardless of the true structure of a given input; that reweighting existing rare examples during training consistently recovers more of the rare classes than undersampling common ones; and that a genuinely balanced training set achieves the best per-class discrimination of any condition tested, at a substantially higher data-generation cost. We conclude with the central open limitation of this approach: because no real burst has a verifiable pulse count, the model's predictions cannot be directly validated against real Swift/BAT data, only checked for consistency against an independent peak-counting proxy in future work.

1. Introduction

Gamma-ray bursts are the most luminous electromagnetic transients known, observed as brief, intense flashes of gamma-ray photons lasting from milliseconds to several minutes. Rather than a single smooth flash, GRB light curves — photon count rate as a function of time — typically show a highly variable, multi-peaked structure. Each distinguishable pulse in this structure is thought to correspond to a discrete episode of energy release at the central engine, such as a distinct accretion episode or a magnetic reconnection event within the relativistic jet (Stern & Svensson 1996). The number and arrangement of these pulses therefore carries information about the central engine's activity pattern, not merely its total output — but the central engine itself is never observed directly; it can only be inferred from the light curve's structure.

In practice, recovering that structure is difficult. Once more than a handful of pulses are present, their rise-and-decay profiles overlap, and faint or low-signal features become hard to separate from instrumental noise. Manual counting is subjective and does not scale; algorithmic peak-detection methods (e.g., MEPSA) impose a fixed detection threshold and so systematically miss overlapping or sub-threshold pulses, recovering a related but distinct quantity — the number of *observable* peaks above some significance level — rather than the *total* number of pulses the underlying process actually produced.

This work reframes pulse counting as a supervised learning problem. Rather than attempting to count pulses in real, unlabeled data, we generate synthetic light curves from a validated stochastic model for which the true pulse count is known by construction, and train a 1D CNN to recover that count directly from the light curve's time series. This sidesteps the central obstacle to any data-driven approach to this problem: real bursts have no verifiable ground-truth pulse count to supervise against in the first place, since "number of parent pulses" is a property of a generative model's internal bookkeeping, not something any instrument measures.

2. Related Work

Stern & Svensson (1996) proposed that GRB light curves can be modeled as the output of a stochastic pulse-avalanche process: a small number of independent "parent" pulses are drawn at the start of a burst, each of which may recursively trigger "child" pulses, producing the characteristic multi-peaked, self-similar structure observed in real bursts. The model has seven free parameters governing the number of parent pulses, the branching behavior of child pulses, and the timing and shape of individual pulses.

Bazzanini et al. (2024) fit this model's seven parameters against real GRB catalogs using a genetic algorithm, producing two independently calibrated parameter sets — one for the BATSE instrument and one for Swift/BAT — such that light curves simulated under a given calibration are statistically consistent with that instrument's real observed burst population. The present work uses the Swift/BAT-calibrated parameter set directly; the genetic-algorithm fitting procedure itself is not rerun here.

Algorithmic peak-detection methods such as MEPSA represent the conventional alternative to a learned approach: they identify statistically significant local maxima in a light curve above a detection threshold. Such methods answer a related but different question from the one addressed here — how many peaks are individually resolvable above the noise — rather than how many pulses the underlying generative process actually produced, some of which may overlap or fall below any fixed significance threshold.

3. Data and Methods

3.1 Stochastic Pulse-Avalanche Model

Following Stern & Svensson (1996), the number of parent pulses in a given light curve, `n_parent_pulses`, is drawn once per light curve from a Poisson distribution with mean `mu0`: `n_parent_pulses ~ Poisson(mu0)`, redrawn if the sampled value is zero. Each parent pulse may recursively trigger child pulses, with the number of children per pulse drawn from a separate Poisson distribution with mean `mu`; `mu ≈ 1` places this branching process near a critical regime, capable of producing deep recursive cascades, bounded in practice by a fixed cap on the total number of pulses per light curve. The remaining model parameters (`alpha, delta1, delta2, tau_min, tau_max`) govern the amplitude and temporal shape of individual pulses. All parameters other than `mu0` are held fixed throughout this work at the Bazzanini et al. (2024) Swift/BAT-calibrated values (`mu=1.34, alpha=2.53, delta1=-0.75, delta2=0.27, tau_min=0.03, tau_max=56.8`); `mu0` is the only parameter varied, as described below. Simulated light curves are passed through a model of the Swift/BAT instrument response (0.064s time resolution, 1,400 cm² effective area, a calibrated background rate, and detection thresholds on burst duration and signal-to-noise) before being retained as training examples, so that retained light curves are consistent with what Swift/BAT would actually have detected and cataloged.

3.2 Synthetic Dataset Generation

Three label-distribution regimes were generated, at two dataset sizes each (N = 3,000 and N = 10,000 light curves):

- **Natural (μ0 = 1.16)** — the Bazzanini et al. (2024) Swift/BAT-fitted value, producing a heavily right-skewed label distribution in which 77-86% of light curves have only 1-2 parent pulses.
- **μ0 = 10** — a deliberately shifted mean, producing a broader, substantially more symmetric label distribution centered near 9-11 pulses (a Poisson distribution's skewness scales as `1/√μ0`, so larger `μ0` produces a visibly more Gaussian-shaped label distribution as a direct consequence of this scaling, independent of any astrophysical assumption).
- **Uniform-by-construction** — rather than a single fixed `μ0`, this dataset was built by sweeping `μ0` across each target pulse-count bin in turn and retaining only light curves whose realized pulse count matched that bin exactly, yielding a genuinely flat label distribution by design rather than by resampling. The computational cost of this method grows sharply with target bin, so this regime was capped at bins 1-8 (N=3,000) and 1-7 (N=10,000) rather than the full range observed under μ0=10.

Table 1. Composition of every dataset generated for this study.

Dataset	N	True labels present	Notes
Natural, N=3,000	3,000	1-6	77-86% of examples are 1-2 parent pulses
Natural, N=10,000	10,000	1-8	same shape as N=3,000, more examples per bin

Dataset	N	True labels present	Notes
μ0=10, N=3,000	3,000	2-25	bin 1 absent — P(X=1|λ=10) ≈ 0.045%, expected count ≈ 1.4, absence consistent with chance
μ0=10, N=10,000	10,000	1-23	
Uniform, N=3,000	2,400	1-8	exactly 300 examples per bin by construction
Uniform, N=10,000	7,000	1-7	exactly 1,000 examples per bin; bin 8 omitted (cost, §3.2)

3.3 Network Architecture

Light curves (peak-normalized to their own maximum, 15,783 time-bins at Swift/BAT's native 0.064s resolution) are passed through a compact 1D CNN, `PulseCountCNN`, summarized layer-by-layer in Table 2.

Table 2. Network architecture. `k` =kernel size, `s` =stride, `p` =padding, in samples.

Stage	Layers	Output channels	Output length
Input	—	1	15,783
Block 1	Conv1d(k=15, s=2, p=7) → BatchNorm1d → ReLU → MaxPool1d(4)	16	1,973
Block 2	Conv1d(k=9, s=2, p=4) → BatchNorm1d → ReLU → MaxPool1d(4)	32	246
Block 3	Conv1d(k=5, s=2, p=2) → BatchNorm1d → ReLU	64	123
Pooling	AdaptiveAvgPool1d(1)	64	1
Head	Flatten → Linear(64→32) → ReLU → Dropout(p=0.2) → Linear(32→1)	1	—

Kernel size decreases with depth (15 → 9 → 5 samples) while channel count increases (16 → 32 → 64), a standard convolutional taper: early layers use a wide window to detect a pulse's rise-and-decay shape directly in the raw signal (a kernel of 15 samples at 0.064s/sample spans roughly 1 second), while deeper layers operate on an already-downsampled representation and combine simpler features into more abstract ones. Each block's combination of a stride-2 convolution and, in the first two blocks, 4-sample max-pooling shrinks the sequence length by roughly 8x per block while growing the receptive field substantially faster than the kernel size alone would suggest — by Block 3, a single output position already integrates information from roughly 389 samples of the original input, and the final global average pool integrates over the entire remaining sequence, giving the network an effectively global view of the light curve despite being built entirely from local convolutions. Average, rather than max, pooling was used for this final step specifically because the quantity being predicted is a count: averaging a channel's activity over time yields a

value that scales with how often a detected pattern recurred, whereas max-pooling would only indicate whether it occurred at least once, discarding exactly the frequency information the task requires. Global average pooling also makes the architecture's total parameter count independent of input sequence length. The output is passed through `softplus(x) + 1`, encoding the constraint `n_parent_pulses ≥ 1` directly in the architecture. The same architecture was used, trained independently from scratch with no shared weights, for every one of the 28 conditions reported below.

3.4 Training Configurations

For the two fixed-`μ0` regimes, three strategies for handling label imbalance were compared, all applied at training time to one shared, already-generated dataset:

- **Unweighted (`unw`)** — the baseline: ordinary random sampling over the dataset exactly as generated, with no correction for class frequency.
- **Weighted (`wgt`)** — training batches are drawn via a weighted random sampler using inverse-class-frequency weights, so that under-represented labels are shown to the model more often per epoch than their raw frequency in the dataset would imply. No examples are discarded; rare examples are simply resampled with replacement more often during training.
- **Capped (`cap`)** — before training, any label value with more than a fixed threshold (1,000) of examples is randomly undersampled down to that threshold; labels already at or below the threshold are left untouched. Sampling during training is then unweighted on this smaller, flatter dataset. Unlike weighting, this strategy permanently discards majority-class data rather than reweighting how it is seen.

The uniform-by-construction regime required none of these, since its labels are balanced by generation rather than by any training-time correction. Each condition was additionally trained under two loss functions — mean squared error (MSE), and Poisson negative log-likelihood, motivated by `n_parent_pulses` itself being Poisson-distributed by construction — giving a 3 (imbalance strategy) × 2 (loss) factorial for each fixed-`μ0` regime and a loss-only comparison for the uniform regime. In total, 28 model instances were trained across both dataset sizes and all regime/strategy/loss combinations (MSE was omitted for `μ0=10` at N=10,000 in an initial pass and completed subsequently; all combinations are reported here).

3.5 Evaluation Metrics

Each trained model was evaluated on a held-out test split (15% of its dataset) using mean absolute error (MAE), exact-match accuracy, and within-±1 accuracy (predictions off by at most one pulse), alongside per-class accuracy broken out by true pulse count.

4. Results

4.1 Naive training collapses onto the dominant label

Under the natural (μ0=1.16) regime, unweighted training reproduces the label distribution's shape in its predictions: exact-match accuracy is highest on classes 1-2 (which dominate the training data) and falls to near zero for classes 3 and above, regardless of dataset size. This produces a deceptively strong aggregate score — unweighted models top both exact-match and within-±1 accuracy in Tables 3 and 4 (§4.5) — that is an artifact of the label distribution rather than evidence of a better model: guessing near the mode is correct most of the time when the mode dominates the test set as heavily as it does here.

4.2 Reweighting outperforms undersampling

Weighted sampling (rare examples shown more often during training, without discarding data) reaches classes that unweighted or capped training never engages at all. At N=3,000, weighted MSE training recovers 21.9% and 4.3% exact-match accuracy on classes 3 and 4 respectively, versus 0% for unweighted and capped training on both classes; this pattern holds at N=10,000 and, as detailed below, under the μ0=10 regime as well. Capped (undersampling) training only produces comparable gains when undersampling happens to tie several classes at exactly equal size — at N=3,000 under the natural regime, for instance, where classes 1-2 are undersampled to the same size, some genuine discrimination between them emerges; at N=3,000 under μ0=10, where no class exceeds the undersampling threshold, capping is a complete no-op, numerically identical to unweighted training. The same undersampling threshold becomes genuinely active at N=10,000 under μ0=10 (several classes there naturally exceed it), but even so, it only redistributes accuracy within the already-populous classes it directly touches — it does not reach classes that were rare to begin with, since by construction it can only remove examples from classes *above* its threshold. Reweighting is the only strategy tested here that reaches genuinely rare classes at all, in every regime and at both dataset sizes tested.

4.3 Loss function choice interacts with resampling strategy

Contrary to the theoretical expectation that Poisson negative log-likelihood should better match Poisson-distributed count labels, MSE weighted training modestly outperformed Poisson weighted training on rare-class recovery under the natural regime at both dataset sizes (e.g., class 3 exact-match of 28.6% vs. 18.5% at N=10,000). Under μ0=10, the two losses perform comparably once weighted sampling is applied (within-±1 accuracy of 33.9% for MSE versus 35.3% for Poisson at N=10,000), suggesting the sampler, not the loss function, is doing most of the work of reaching rare classes; why Poisson loss underperforms its theoretical motivation specifically when paired with aggressive resampling is not resolved by this work. The one regime where Poisson loss shows a clear, uncontested advantage is the uniform-by-construction dataset at N=3,000 (within-±1 accuracy of 45.8% vs. 40.6% for MSE) — the only condition in this study with no competing resampling mechanism to interact with the loss function's own behavior. That advantage narrows sharply at N=10,000 (55.6% vs. 54.5%), consistent with a general expection that larger training sets reduce sensitivity to loss-function choice.

4.4 A genuinely balanced dataset outperforms every resampling strategy

The uniform-by-construction regime achieves the best per-class discrimination of any condition tested at a given dataset size — its within-±1 accuracy (45.8-55.6% across sizes) exceeds every weighted or capped condition under the natural or μ0=10 regimes on the same metric, though the comparison is not perfectly like-for-like, since the uniform regime's label range (1-7/1-8) is narrower than the other two regimes' (1-6+ and 2-25 respectively) — a narrower label range makes within-±1 accuracy an easier target independent of model quality. Within that caveat, the uniform regime is the only one in which every tested label value receives meaningfully engaged predictions rather than a small subset of "reachable" classes, which we attribute to there being no structural imbalance left for any training-time strategy to correct or fail to correct.

4.5 Full results, all conditions

Tables 3 and 4 report aggregate test-set metrics for all 14 conditions trained at each dataset size. Read with the caveat established in §4.1: the unweighted rows' high aggregate scores reflect the concentration of the label distribution, not superior per-class modeling; per-class behavior is discussed in §4.1-§4.4.

Table 3. All conditions, N=3,000.

Regime	Condition	Test MAE	Exact-match	Within-±1
Natural (μ0=1.16)	MSE, unweighted	0.633	46.2%	92.2%
Natural (μ0=1.16)	MSE, weighted	0.758	34.4%	91.3%
Natural (μ0=1.16)	MSE, capped	0.720	37.6%	92.3%
Natural (μ0=1.16)	Poisson, unweighted	0.658	44.4%	91.6%
Natural (μ0=1.16)	Poisson, weighted	0.749	42.7%	86.0%
Natural (μ0=1.16)	Poisson, capped	0.691	40.6%	92.3%
μ0=10	MSE, unweighted	2.429	10.9%	34.2%
μ0=10	MSE, weighted	2.802	10.0%	33.1%
μ0=10	MSE, capped	2.429	10.9%	34.2%
μ0=10	Poisson, unweighted	2.427	12.0%	36.4%
μ0=10	Poisson, weighted	2.793	10.4%	32.4%
μ0=10	Poisson, capped	2.427	12.0%	36.4%
Uniform-by-construction	MSE	1.861	13.9%	40.6%
Uniform-by-construction	Poisson	1.731	15.0%	45.8%

Table 4. All conditions, N=10,000.

Regime	Condition	Test MAE	Exact-match	Within-±1
Natural (μ0=1.16)	MSE, unweighted	0.670	43.7%	91.6%
Natural (μ0=1.16)	MSE, weighted	0.801	35.9%	86.5%

Regime	Condition	Test MAE	Exact-match	Within-±1
Natural (μ0=1.16)	MSE, capped	0.860	32.6%	83.2%
Natural (μ0=1.16)	Poisson, unweighted	0.660	45.0%	91.8%
Natural (μ0=1.16)	Poisson, weighted	0.750	39.5%	87.9%
Natural (μ0=1.16)	Poisson, capped	0.843	35.7%	83.1%
μ0=10	MSE, unweighted	2.291	12.7%	39.3%
μ0=10	MSE, weighted	2.715	11.6%	33.9%
μ0=10	MSE, capped	2.391	12.4%	37.6%
μ0=10	Poisson, unweighted	2.297	13.1%	39.9%
μ0=10	Poisson, weighted	2.631	12.3%	35.3%
μ0=10	Poisson, capped	2.444	11.9%	37.0%
Uniform-by-construction	MSE	1.475	20.3%	54.5%
Uniform-by-construction	Poisson	1.438	19.8%	55.6%

Notably, the `capped` condition is numerically identical to `unweighted` under μ0=10 at N=3,000 for both losses (0.9-2.427 MAE match to three decimal places) — the undersampling threshold (1,000) never triggers at this size, since the largest class there has only 374 examples, so no data is ever actually removed. The same threshold becomes genuinely active at N=10,000, where several classes naturally exceed it, and `capped` there is no longer identical to `unweighted` (Table 4).

4.6 Best-performing condition per dataset

Table 5 summarizes the best-performing training condition for each dataset regime and size, selected for genuine per-class engagement rather than aggregate score alone (§4.1).

Table 5. Best-performing condition per dataset (by within-±1 accuracy among conditions that engage more than the dominant class alone).

Dataset	Best condition	Test MAE	Exact-match	Within-±1
μ_0 = 1.16, N=3,000	MSE, weighted	0.758	34.4%	91.3%
μ_0 = 1.16, N=10,000	MSE, weighted	0.801	35.9%	86.5%
μ_0 = 10, N=3,000	Poisson, weighted	2.793	10.4%	32.4%
μ_0 = 10, N=10,000	Poisson, weighted	2.631	12.3%	35.3%
Uniform, N=3,000	Poisson	1.731	15.0%	45.8%
Uniform, N=10,000	Poisson	1.438	19.8%	55.6%

Within-±1 accuracy is not directly comparable down this column, since it depends on the width of each regime's true label range, not solely on model quality; the natural regime's high values (83-92%) reflect labels concentrated on 1-2 pulses, not a categorically better model than the μ0=10 regime's 32-35% over a much wider label range.

5. Discussion

The consistent finding across every regime and dataset size tested is that reweighting how existing examples are sampled during training reaches further into a label distribution's rare tail than any strategy that resamples or discards existing data, and that this advantage holds regardless of loss function. Undersampling (capping) only ever produces comparable gains as a side effect of coincidentally tying multiple classes at exactly equal size after thinning — a structural accident of the specific undersampling threshold chosen relative to the specific class sizes present, not a general property of undersampling as a strategy. This suggests that, for problems structurally similar to this one — a single dominant class with a long, sparse tail — reweighting during training is the more robust default remedy, with undersampling better reserved for cases where the practitioner can verify in advance that it will tie enough classes together to be worthwhile.

The uniform-by-construction dataset's superior per-class discrimination, at the cost of a substantially higher data-generation budget, points to a broader implication: where feasible, correcting label imbalance at the data-generation stage dominates any correction applied at training time, because training-time corrections operate only on the data that already exists, while generation-time correction can supply genuinely new, appropriately distributed examples of classes that would otherwise remain rare no matter how they are resampled.

5.1 Limitations

The central limitation of this work is that its central claim — accurate pulse counting — cannot be directly verified against real observational data. `n_parent_pulses` is a property of the generative model used to create the training data, not a directly observable quantity; no real Swift/BAT burst carries a verified ground-truth pulse count to check a prediction against, even in principle. A weaker proxy validation is possible in future work: comparing this model's predictions on real Swift/BAT light curves against an independent significant-peak count from an algorithmic method such as MEPSA. Because MEPSA counts only peaks above a fixed detection threshold while this model is trained to recover the full generative pulse count (including overlapping or sub-threshold pulses), agreement or disagreement between the two would not constitute definitive validation, only a qualitative consistency check; this comparison was not performed in the present work due to lack of access to the real catalog and precomputed peak-detection outputs. Further limitations include: a single fixed instrument/parameter calibration (Swift/BAT) with no tested generalization to BATSE or other instrument calibrations; single train/validation/test splits per condition without repeated-seed averaging; and the uniform-by- construction dataset's bin range being capped below its originally intended extent due to the steeply rising computational cost of harvesting high-pulse-count examples by this method.

6. Conclusion

A 1D CNN can recover the pulse-avalanche model's parent-pulse count from a synthetic GRB light curve's time series, but naive training on an imbalanced label distribution collapses onto the

dominant class regardless of the specific input, an effect that persists across two independently generated label-distribution regimes and two dataset sizes. Reweighting existing rare examples during training consistently recovers more of the label distribution's rare tail than undersampling common classes, independent of loss function; a genuinely balanced training set, built by construction rather than by resampling, achieves the best per-class discrimination of any condition tested, at a significant increase in data-generation cost. Because the target label is unobservable in real data, this model's predictions remain to be checked for consistency against real Swift/BAT bursts using an independent peak-counting method as a proxy, which is the natural next step for this work.

References

Stern, B. E. & Svensson, R. 1996, *Astrophysical Journal Letters*, 469, L109, "Evidence for Chain Reaction in the Time Profiles of Gamma-Ray Bursts."

Bazzanini, L., Ferro, L., Guidorzi, C., Angora, G., Amati, L., Brescia, M., Bulla, M., Frontera, F., Maccary, R., Maistrello, M., Rosati, P., & Tsvetkova, A. 2024, *Astronomy & Astrophysics*, 689, A266, "Long gamma-ray burst light curves as the result of a common stochastic pulse-avalanche process." doi:10.1051/0004-6361/202450150